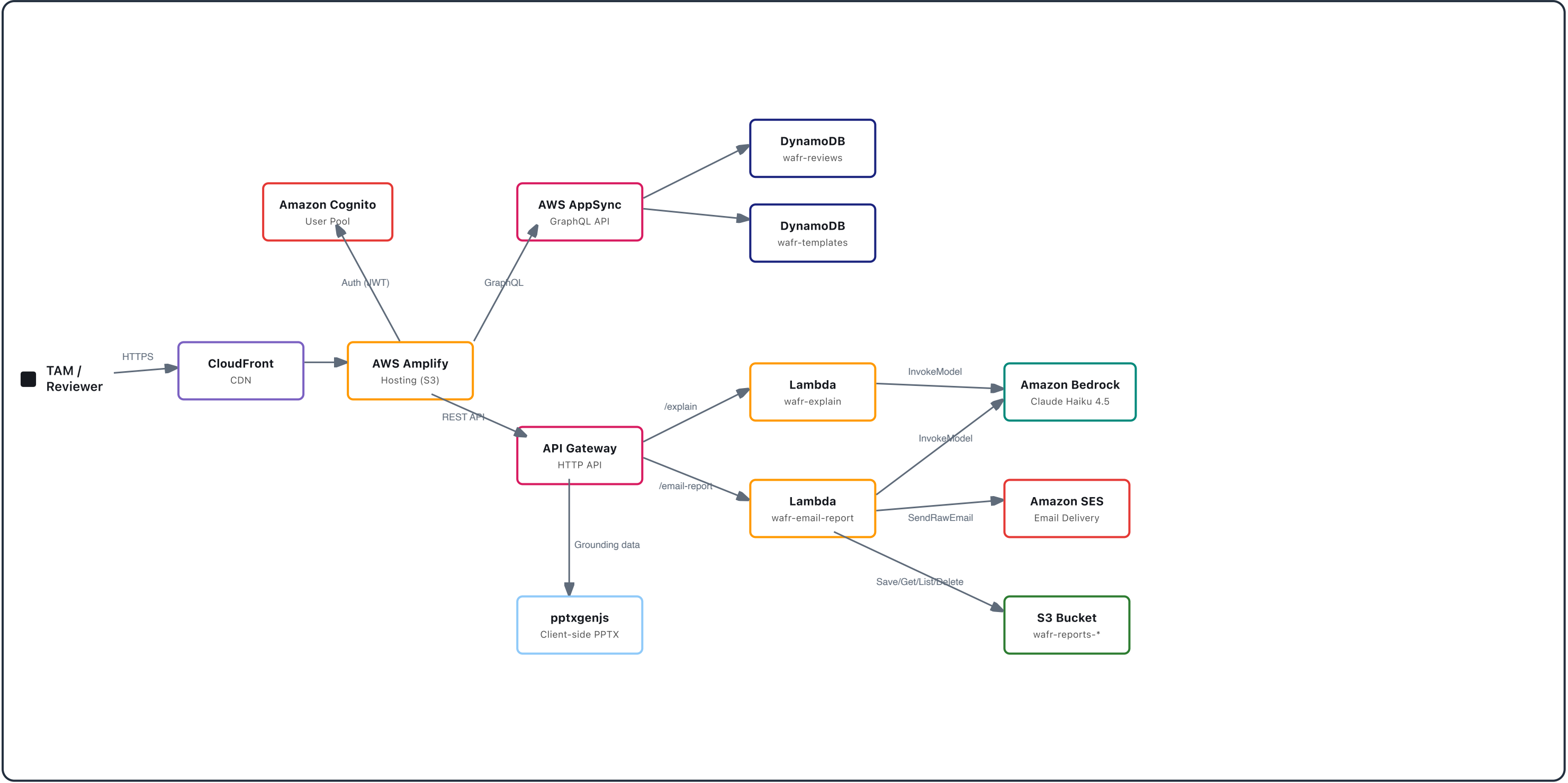


AWS WorkSpaces WAFR Tool — Architecture

AWS Cloud (Account: 590183747733 — eu-west-2)



Data Flows & Lambda Actions

Flow	Route / Action	Services	Description
Authentication	Cognito SRP	Cognito → Browser (JWT)	User signs in, JWT stored in localStorage for API calls

Flow	Route / Action	Services	Description
Cloud Sync	GraphQL mutations/queries	AppSync → DynamoDB	CRUD operations on wafr-reviews and wafr-templates
AI Guidance	POST /explain	API GW → Lambda → Bedrock	Concise TAM guidance per question (512 tokens)
Report Generation	POST /email-report action: generate	API GW → Lambda → Bedrock	Batched (5 Qs/call), parallel Bedrock calls, 4096 tokens
Email Report	POST /email-report action: email	API GW → Lambda → SES	HTML report as email attachment via SES
Save Report	POST /email-report action: saveReport	API GW → Lambda → S3	Auto-saves Standard/So What/C-Level reports to S3
List Reports	POST /email-report action: listReports	API GW → Lambda → S3	Lists saved reports for a review (by prefix)
Get Report	POST /email-report action: getReport	API GW → Lambda → S3	Returns base64 report content (direct GetObject)
Delete Report	POST /email-report action: deleteReport	API GW → Lambda → S3	Removes saved report from S3
Curate C-Level Deck	POST /email-report action: curateDeck	API GW → Lambda → Bedrock	AI curates findings into executive tiers (8192 tokens)
Email Notification	POST /email-report action: notify	API GW → Lambda → SES	Fire-and-forget notification (deck complete/failed)
AI Chat	POST /email-report action: chat	API GW → Lambda → Bedrock	Grounded Q&A on review data + saved reports (1024 tokens)
C-Level PPTX	Client-side (pptxgenjs)	Browser → S3 (via saveReport)	13-slide deck built in browser, saved to S3

AWS Services

Service	Resource	Purpose	Config
Amplify Hosting	App: d1p2543h8l2mfc	Static site hosting + CDN	Manual zip deploy
CloudFront	Auto (via Amplify)	Content delivery	—
Cognito	User Pool: eu-west-2_Wy0eJHyN3	User authentication (SRP + JWT)	2 users
AppSync	API: 4up36qgqubd6tcuekx5cmexmii	GraphQL API for reviews + templates	Cognito auth
DynamoDB	wafr-reviews	Review storage	PAY_PER_REQUEST, PITR
DynamoDB	wafr-templates	Template storage	PAY_PER_REQUEST, PITR
API Gateway	HTTP API: 6ylrfwa3d8	REST endpoints (/explain, /email-report)	CORS enabled
Lambda	wafr-explain	AI guidance per question	Python 3.12, 30s, 128MB
Lambda	wafr-email-report	Reports, email, S3 ops, chat, curateDeck, notify	Python 3.12, 90s, 256MB
Bedrock	eu.anthropic.claude-haiku-4-5-20251001-v1:0	AI explanations, recommendations, curation, chat	EU inference profile
SES	Sender: danmmat@amazon.co.uk	Email delivery for reports + notifications	Sandbox mode
S3	wafr-reports-danmmat-9219112	Auto-saved reports (HTML + PPTX)	eu-west-2

Monthly Cost Estimate

Service	Expected Usage	Cost
Amplify Hosting	<1GB served	Free tier
DynamoDB (+ PITR)	<25 WCU/RCU, <1GB	~\$0.20
AppSync	<250K queries	Free tier
Cognito	1-2 users	Free tier
API Gateway	<100 requests/month	Free tier
Lambda	<150 invocations/month	Free tier

Service	Expected Usage	Cost
Bedrock (Haiku 4.5)	~100-200 calls (512-8192 tokens)	~\$0.20
S3 (Reports)	<1GB stored, <200 requests	~\$0.03
SES	<50 emails/month	Free tier
Total		<\$1/month

Last updated: July 2025 | Diagram version aligned with ARCHITECTURE.md